

Fusion ERP

**Software
Requirement
Specification
(SRS)
for
Patent
Management
System Module**

1. Introduction

1.1 Overview of the Fusion System

FusionIIIT is a comprehensive platform developed by students at PDPM Indian Institute of Information Technology, Design, and Manufacturing, Jabalpur, using Python 3.8 and the Django web framework. Fusion serves as a unified system designed to streamline administrative, academic, and departmental activities. By integrating various modules, it enhances efficiency, simplifies processes, and ensures smooth campus-wide operations.

1.2 Purpose of the Patent Management System

The Patent Management System module's primary objective is to streamline the patent filing process within the institute. It provides a structured platform where faculty members and students can file Intellectual Property (IP) forms, track their submissions, and interact with key stakeholders such as the PCC Admin and attorneys. The module automates the review, approval, and patent filing process, reducing manual effort and improving coordination.

1.3 Scope of the Patent Management System

This module is focused on simplifying the end-to-end process of patent management. It allows users to:

- Submit patent filing forms.
- Track the status of applications.
- Facilitate communication between attorneys and applicants for modifications.
- Enable final submission to relevant authorities for patent filing.

The module needs to be designed with robust security features to handle sensitive information and provides different levels of access based on user roles, ensuring that only authorised personnel can view or modify specific documents.

2. Overall Description:

2.1 User/Actor Characteristics

2.1.1 Applicant

Role: Submitters of patent applications.

Responsibilities:

- Fill out the Intellectual Property Filing Form.
- Submit the form for approval.
- Track the progress of the filing.
- Gets notified about status of application

2.1.2 PCC Admin

Role: Manager of patent applications.

Responsibilities:

- Review submitted patent applications.
- Approve or request changes to the applications.
- Forward approved applications to the Director for approval and then to attorneys for further review.

2.1.3 Director

Role: Approver of patent applications.

Responsibilities:

- Review patent applications forwarded by the PCC Office
- Approve or reject patent applications based on the review.
- Provide feedback or request changes if the application is rejected.
- Review the attorney's patentability confirmation (design or product patent).
- Give final approval after attorney confirmation for contract creation.
- Ensure that all patent applications follow legal and institutional guidelines.

2.2 Dependencies

2.2.1 UI-Level Dependencies

- **Dashboard Integration:** The patent management system should integrate smoothly with the existing dashboard, allowing users to navigate between the patent system and other modules.
- **Notification Integration:** Users should be notified of important updates (such as application status changes) through real-time notifications, which will be displayed on the central dashboard.

2.2.2 Database-Level Dependencies

- **User and Document Models:** The database must support structured models for storing user data (roles, profiles) and patent documents. These models must be optimised for fast retrieval and efficient storage.
- **Document Storage:** Patent documents (e.g., descriptions, and designs) must be stored securely and linked to user profiles for easy access and updates.

2.2.3 Module-Level Dependencies

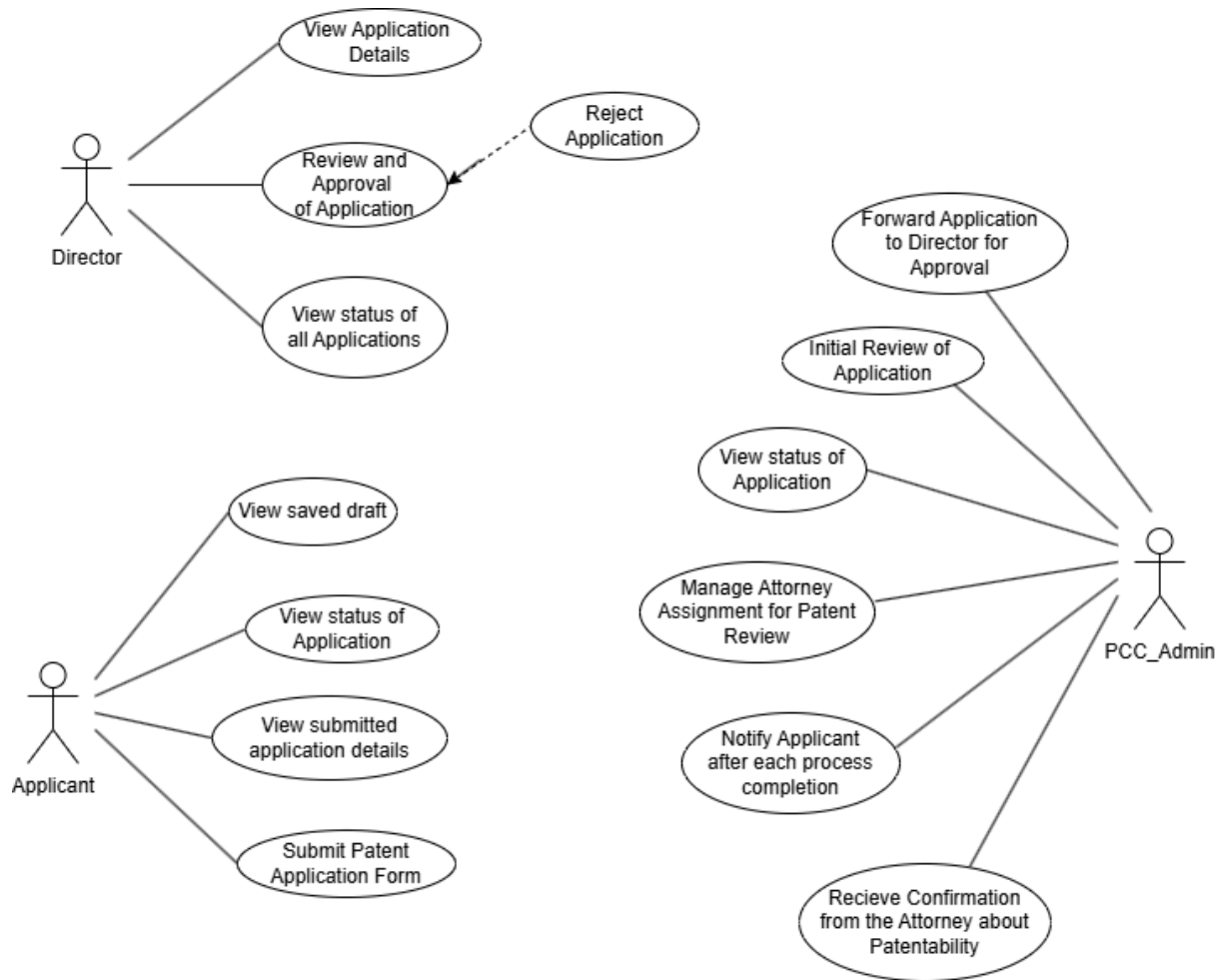
- **User Management:** This module ensures that user roles and access rights are properly assigned, and only authorised individuals can perform specific actions (e.g., patent approval).
- **Document Management:** This module allows users to upload, update, and manage patent-related documents. It also tracks revisions and approval status.
- **Notification System:** Integration with the notifications system ensures users receive timely updates on application progress or important deadlines.
- **Workflow Management:** The system should include customizable workflows to handle different stages of the patent application process, from submission to approval or rejection, allowing for efficient tracking and management of applications.

2.3 Definitions, Acronyms, and Abbreviations

- **PCC:** Patent Copyright Cell.
 - **IP:** Intellectual Property.
 - **Applicant:** Faculty member or student applying for a patent.
 - **Attorney:** Legal personnel responsible for patentability checks.
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3. System Features (Functional Requirements)

3.1. Use Case Diagram:



3.2. Use Case Description:

Use Case 1: Submit Patent Application Form

Use Case ID	1
Use Case Name	Submit Patent Application Form
Description	The applicant submits the patent application form for review.
Actor	Applicant
Precondition	The applicant must be logged in.
Main Flow	1. The applicant logs into the system. 2. Fill out the patent application form. 3. Submit the form for review.
Postconditions	The patent application is submitted and sent for review to the PCC Admin.
Alternate Flow	AF1: Incomplete Application

	1. If the applicant attempts to submit an incomplete application, the system alerts them. 2. The applicant is guided to the missing fields and must complete them before submitting. AF2: Save as Draft 1. The applicant chooses to save the application as a draft. 2. The system saves the incomplete application and allows the applicant to return and submit it later. AF3: Application Timeout 1. If the applicant is inactive for too long, the system logs them out for security purposes. 2. Any unsaved data is lost, and the applicant is notified to log back in to continue the application process.
Sub Flow	SF1: Auto-Validation 1. Before submission, the system performs an automatic validation to ensure required fields are filled out correctly. 2. If validation fails, the system displays error messages for incorrect or missing fields. SF2: Confirmation Page 1. After submitting the form, the system shows a confirmation page. 2. The applicant receives a reference number for their application and confirmation of successful submission. SF3: Notify PCC Admin 1. Once the form is submitted, the system notifies the PCC Admin for review. 2. The PCC Admin receives an email or dashboard notification with the application details.
Global Alternate Flow	GAF1: Cancel Submission 1. The applicant decides to cancel the submission process at any stage before the final submission. 2. The system cancels the process and returns the applicant to the dashboard or previous page. 3. Any unsaved changes are lost, and the applicant can choose to restart or exit.

Use Case 2: View the Status of the Application

Use Case ID	2
Use Case Name	View the Status of the Application
Description	Users can view the current status of their patent application.
Actor	Applicant, PCC_Admin
Precondition	The user must be logged in.
Main Flow	1. The user logs into the system. 2. Navigate to the "View Status" section. 3. View the current status of the application.

Postconditions	The status of the application is displayed to the user.
Alternate Flow	AF1: No Application Submitted 1. If the user has not submitted any application, the system alerts them that there are no applications to view. AF2: Network Issue 1. If the network connection fails while trying to fetch the status, the system displays an error message. 2. The system retries fetching the data, or the user can try again later. AF3: Application Rejected 1. If the application is rejected, the system displays the rejection reason with additional instructions (if any) for appeal or resubmission.
Sub Flow	SF1: Filter Applications by Status 1. The system allows the user to filter applications based on different status categories such as "Pending," "Under Review," "Approved," and "Rejected." 2. The user can select a category and view only those applications. SF2: Notify Applicant on Status Update 1. When there's an update to the application status, the system sends an automatic notification (email or dashboard alert) to the applicant. 2. The notification includes the new status and a link to view more details. SF3: Application History View 1. The system displays a history of status changes (e.g., "Submitted on Date X," "Under Review on Date Y," "Approved on Date Z"). 2. The applicant can view the timeline of the application's progress.
Global Alternate Flow	GAF1: System Maintenance 1. If the system is under maintenance while the user tries to access the status page, the system displays a message about maintenance. 2. The user is notified to check back later when the system is online again. GAF5: Withdraw Application 1. The user decides to withdraw their application after viewing its status.

	2. The system prompts the user to confirm, and if confirmed, the application status is changed to "Withdrawn" and logged in the system.
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Use Case 3: View Attorney Assigned and their Details

Use Case ID	3
Use Case Name	View Attorney Assigned
Description	This use case allows the applicant to view details of the attorney assigned to their patent application. Once an attorney is allocated by the PCC office, the applicant can log into the system and access the attorney's information, which includes the attorney's name, contact details, and any other relevant information needed for communication.
Actor	Applicant
Precondition	1.The applicant must be logged into the system. 2.An attorney must already be assigned to the applicant's patent application.
Main Flow	1.Applicant Logs into the System - The applicant logs in using their valid credentials (username and password). 2.Navigates to "View Attorney Assignment" Page - From the dashboard, the applicant selects the "View Attorney Assignment" option. 3.View Assigned Attorney Details - The system displays the attorney's information, including: - Attorney's name - Contact details (email, phone number) - Attorney's field of expertise (e.g., design patents, product patents) - Any additional instructions provided by the PCC office. 4.Completion - The applicant views the attorney details and can initiate communication as necessary.
Postconditions	1.The applicant successfully views the assigned attorney's details. 2.The applicant may note or save the attorney's contact information for further communication.
Alternate Flow	AF-1: Attorney Not Yet Assigned Precondition: An attorney has not yet been assigned.

	1. The applicant logs into the system and navigates to "View Attorney Assignment." 2. The system displays a message: "No attorney has been assigned to your application yet." 3. The applicant may choose to return to the dashboard or check back later. Postconditions: • The system informs the applicant that no attorney is assigned yet. AF-2: Applicant Has Multiple Pending Applications Precondition: The applicant has submitted multiple patent applications, each with different statuses. 1. The applicant logs into the system and navigates to the "View Attorney Assignment" page. 2. The system detects multiple active patent applications under the applicant's name. 3. The system displays a list of all active applications with corresponding attorney assignments. 4. The applicant selects a specific application to view the attorney details. Postconditions: • The applicant views the attorney assigned for the selected application. • The applicant can switch between multiple applications and their respective attorney details. AF-3: Attorney Information Update by PCC Office Precondition: The PCC office reassigns or updates the attorney details for the applicant's patent application (e.g., due to workload adjustments). 1. The applicant logs into the system and navigates to "View Attorney Assignment." 2. The system displays an updated attorney assignment message: "Your assigned attorney has been changed." 3. The updated attorney details are shown (new contact info, name, expertise, etc.). Postconditions: • The applicant views the newly assigned attorney's details. • The system highlights the change in assignment to ensure the applicant is informed.
Sub Flow	SF-1: Attorney Details Unavailable Precondition: Attorney details are unavailable due to system maintenance or data update issues. 1. The applicant navigates to "View Attorney Assignment."

	- The system encounters an error in fetching attorney details. - A message is displayed: "Attorney details are currently unavailable. Please try again later." Postconditions: - The applicant is informed of the temporary unavailability of attorney details. - The applicant can retry later or contact the PCC office for assistance. SF-2: Applicant Cannot Access the "View Attorney Assignment" Page Precondition: The applicant attempts to access the attorney assignment page but faces an access restriction or permissions issue. - The applicant logs into the system and tries to navigate to the "View Attorney Assignment" page. - The system detects a permissions issue and displays a message: "You do not have access to this feature at the moment." - The applicant can contact support or the PCC office to resolve the issue. Postconditions: - The applicant is unable to view attorney details due to access limitations. - The system directs the applicant to resolve the issue by contacting support. SF-3: Technical Error in Loading Attorney Details Precondition: A technical issue prevents the system from displaying the assigned attorney's details (e.g., server timeout, data retrieval error). - The applicant logs into the system and navigates to the "View Attorney Assignment" page. - The system encounters an error when fetching attorney information. - An error message is displayed: "Unable to retrieve attorney details due to a technical issue. Please try again later." - The applicant may choose to refresh the page or try again later. Postconditions: - The system logs the error and notifies support for investigation. - The applicant is informed of the technical issue and cannot view the attorney details immediately.
Global Alternate Flow	GAF-1: Applicant Logs Out without Viewing Attorney Assignment

	1. The applicant logs into the system but does not view the attorney assignment. 2. The applicant logs out of the system without checking the assigned attorney details. Postconditions: • The session is closed, and the applicant has not accessed the attorney information. GAF-2: System Timeout Due to Inactivity Precondition: The applicant logs into the system but remains inactive for a prolonged period, leading to a session timeout. 1. The applicant logs into the system and navigates to the dashboard. 2. The applicant stays inactive for a specified time (e.g., 10 minutes). 3. The system automatically logs out the applicant due to inactivity. 4. The applicant attempts to navigate to the "View Attorney Assignment" page but is prompted to log in again. Postconditions: • The applicant is redirected to the login page due to the session timeout. • After logging back in, the applicant can attempt to view attorney details again. GAF-3: Internet Connectivity Issue Precondition: The applicant experiences a loss of internet connectivity while logged into the system. 1. The applicant logs into the system and navigates to the "View Attorney Assignment" page. 2. The internet connection is lost before the attorney details can be displayed. 3. The system displays a message: "Connection lost. Please check your internet connection and try again." 4. The applicant can retry once the connection is restored. Postconditions: • The applicant is unable to view attorney details until internet connectivity is restored.
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Use Case 4: Modify and Update Application

Use Case ID	10
Use Case Name	Update Application
Description	Allows the applicant to update or modify details of their patent application after submitting it. Communication between the applicant

	and the PCC Admin regarding any changes outside the system (e.g., via email or phone).
Actor	Applicant
Precondition	The applicant is logged into the system, and an initial patent application has already been submitted.
Main Flow	1. The applicant logs into the system and navigates to the "Update Application" section. 2. The system displays the applicant's submitted patent application details. 3. The applicant updates the required details of the application. 4. The system saves the updated information and confirms the changes.
Postconditions	The updated patent application details are reflected in the database, and the applicant can view the updated status of the application.
Alternate Flow	AF1: Requested Changes by PCC Admin 1. The applicant receives a notification about the requested changes after the initial review by the PCC Admin (outside the system communication). 2. The applicant logs in and navigates to the application to view the specific changes requested. 3. The applicant updates the application according to the requested modifications. 4. The system saves the updated application and notifies the applicant that changes have been successfully submitted. AF2: Application Locked During Attorney Review 1. If the application has already been forwarded to the attorney for review, the system does not allow further updates by the applicant. 2. The applicant is notified that the application is locked for updates. 3. If changes are necessary, the applicant must request a modification from the PCC Admin or wait for the attorney's feedback. AF3: Incomplete Application Submission 1. If the applicant attempts to submit an incomplete application (missing required fields or sections), the system alerts the applicant. 2. The applicant is directed to the missing fields and must complete them before proceeding. 3. Once the missing fields are filled, the system allows the applicant to submit the updates. AF4: Deadline Passed for Updates 1. If the applicant tries to update the application after a submission deadline set by the PCC Admin, the system prevents the update. 2. The applicant is informed that no further updates can be made after the deadline unless an exception is granted by the PCC Admin.
Sub Flow	SF1: Save the Application as Draft

	1. If the applicant does not complete the updates in one session, they can save the application as a draft. 2. The system stores the partially updated application as a draft. 3. The applicant can log in later and resume updating the application before submitting the final version. SF2: Notify Applicant of System-Generated Changes 1. If any system-generated changes are made to the application (e.g., auto-correction of certain fields or formats), the system automatically updates those fields. 2. The applicant is notified of the changes made by the system (e.g., auto-formatting of dates or auto-calculated fields). 3. The applicant reviews the system-generated changes and can either accept or modify them. SF3: Validation of Changes 1. After updating the application, the system runs a validation check to ensure that all required fields are correctly filled. 2. If any validation rules are violated (e.g., incorrect data formats, missing mandatory fields), the system displays an error message. 3. The applicant corrects the errors and resubmits the application for saving.
Global Alternate Flow	GA1: The applicant can cancel the update process at any time, which will redirect them back to the dashboard without saving changes. GA2: If a technical error occurs during the update process (e.g., network failure, server issues), the system displays an error message and logs the incident. GA3: If the system detects that multiple users (applicant, PCC Admin) are trying to update the same application simultaneously, it alerts the applicant of potential conflicts and prevents overwriting.

Use Case 5: Initial Review of Application

Use Case ID	5
Use Case Name	Initial Review of the Application
Description	The PCC Admin conducts an initial review of the patent application for completeness.
Actor	PCC_Admin
Precondition	The admin must be logged in. A patent application has been submitted by the applicant.
Main Flow	1. The PCC Admin logs into the system. 2. Review the patent application. 3. Either forward it for approval or request changes from the applicant.
Postconditions	The application is either forwarded to the next stage for approval or sent back to the applicant for updates.

Alternate Flow	If the application is incomplete or incorrect, the PCC Admin requests revisions and sends it back to the applicant.
Sub Flow	The system auto-saves the admin's review notes or comments in case the review is interrupted, allowing the admin to resume later without losing progress.
Global Alternate Flow	The PCC Admin cancels the action and logs out.

Use Case 6: Notify Applicant

Use Case ID	6
Use Case Name	Notify Applicant of changes required.
Description	The PCC Admin notifies the applicant that changes are required in their application.
Actor	PCC Admin
Precondition	The PCC admin must be logged in.
Main Flow	1. The PCC Admin logs into the system. 2. Review the application. 3. Notifies the applicant that changes are needed. 4. The status of the application becomes "Requires Modification".
Postconditions	The applicant is informed of the changes required.
Alternate Flow	AF1: Admin identifies no changes needed If, during the review, the PCC Admin determines that no changes are required in the application: 1. The PCC Admin reviews the application. 2. The PCC Admin decides that no changes are necessary. 3. The Admin approves the application without notifying the applicant. 4. The status of the application is updated to "Approved." AF2: System Error or Network Failure If a system error or network failure occurs while the PCC Admin is trying to notify the applicant: 1. The PCC Admin reviews the application. 2. The Admin attempts to notify the applicant, but the system encounters an error or network failure. 3. The system displays an error message. 4. The PCC Admin retries the notification process or contacts technical support for assistance. 5. Once the system is restored, the Admin can successfully notify the applicant.
Sub Flow	SF 1: Notify Applicant via Different Medium If the system provides multiple channels to notify the applicant

	(e.g., email, SMS, or portal notification), the sub flow for notifying the applicant through different means would be: 1. The PCC Admin selects the method for notifying the applicant (e.g., email, SMS, portal). 2. The system sends the notification to the applicant using the chosen medium. 3. The applicant receives the notification and is informed of the changes required.
Global Alternate Flow	The PCC Admin cancels the action and logs out.

Use Case 7: Forward Application to Director for Approval

Use Case ID	7
Use Case Name	Forward the Application to the director for approval.
Description	The PCC Admin forwards the patent application to the Director for final approval.
Actor	PCC_Admin
Precondition	1. The PCC Admin must be logged in. 2. PCC_Admin must have completed the initial review.
Main Flow	1. The PCC Admin logs into the system. 2. Review the application. 3. Forward the application to the Director for approval.
Postconditions	The application is successfully forwarded to the Director.
Alternate Flow	Invalid Application Found: • If the PCC Admin identifies any discrepancies or incomplete information during the review, the application is returned to the applicant for correction. ○ The PCC Admin selects the “Return for Corrections” option. ○ A notification is sent to the applicant with the required corrections. ○ The applicant makes corrections and resubmits the application. ○ The PCC Admin is notified of the resubmission and resumes the review. System Error/Failure: • If the system experiences a technical issue while attempting to forward the application: ○ The system logs the error. ○ The PCC Admin is presented with an error message and can choose to: ■ Retry the forwarding action. ■ Save the current progress and log out. ○ The system administrator is notified of the failure for troubleshooting.
Sub Flow	Application Comments:

	- The PCC Admin can add comments to the application before forwarding it to the Director. - These comments may highlight specific points of interest, additional notes, or recommendations for the Director to consider during approval. Confirmation Dialog: - Before the application is forwarded, the system prompts the PCC Admin to confirm the action. - The PCC Admin clicks “Confirm” to proceed or “Cancel” to abort the forwarding process.
Global Alternate Flow	The PCC Admin cancels the action and logs out.

Use Case 8: Approve or Reject Patent Application

Use Case ID	8
Use Case Name	Approve or Reject Patent Application
Description	The Director makes a decision to accept or reject the patent application.
Actor	Director
Precondition	- The application has been reviewed by a PCC Admin. - The Director must be logged into the system.
Main Flow	- The Director opens the reviewed application. - The Director clicks "Accept" or "Reject" to make a decision and can write suggestions. - The system records the decision and notifies the PCC_Admin and Applicant.
Postconditions	The patent application is accepted or rejected, and the applicant is notified.
Alternate Flow	If the Director requires more information, they can request additional clarification from the PCC Admin before deciding.
Sub Flow	The Director reviews supporting documents provided by the PCC Admin for further review.
Global Alternate Flow	The Director cancels the action and returns to the dashboard without making a decision.

Use Case 9: Request Modifications from PCC Admin

Use Case ID	9
Use Case Name	Request Modifications from PCC_Admin
Description	The Director requests additional clarifications or supporting documents from the PCC Admin before making a decision.
Actor	Director, PCC Admin

Precondition	The Director has reviewed the application but requires more information.
Main Flow	1. The Director reviews the application. 2. Requests clarifications from the PCC Admin. 3. The PCC Admin responds with additional documents. 4. The Director reviews the new information.
Postconditions	The Director decides after reviewing additional information.
Alternate Flow	None
Sub Flow	None
Global Alternate Flow	The Director cancels the request and returns to the dashboard.

Use Case 10: Manage Attorney Assignment

Use Case ID	10
Use Case Name	Manage Attorney Assignment
Description	The admin manages the attorney assigned to review the applications.
Actor	PCC_Admin
Precondition	The admin must be logged in.
Main Flow	1. The admin logs into the system. 2. Navigate to the "Attorney Assignment" section from the dashboard. 3. The system displays a list of applications that need an attorney assigned or have an attorney already assigned. 4. The admin selects an application to view or modify its attorney assignment. 5. The system displays the current attorney assignment (if any). 6. The admin selects an attorney from the list to assign, reassign, or remove as needed
Postconditions	The admin successfully views the assigned attorney.
Alternate Flow	AF1: Failure to Display Attorney List: • After navigating to "View Attorney Assignment," the system fails to display the attorney list due to a technical issue (e.g., network error, system malfunction). 1. The system shows an error message. 2. The admin can retry or contact support for resolution. AF2: No Attorney Assigned: • After navigating to "View Attorney Assignment," the system indicates that no attorney is currently assigned to the selected application. 1. The system prompts the admin to assign an attorney.

	2. The admin can proceed to assign an attorney to the application
Sub Flow	SF1: Viewing Current Attorney Assignment 1. The admin navigates to "Attorney Assignment." 2. Select an application to view. 3. The system displays the currently assigned attorney (if any) or shows "No attorney assigned."
Global Alternate Flow	The admin logs out without viewing or modifying the assignment.

Use Case 11: View Patent Applications

Use Case ID	11
Use Case Name	View Active Patent Applications
Description	The Director views active patent applications for review or awaiting further action.
Actor	Director
Precondition	The Director must be logged into the system, and active applications must be available.
Main Flow	1. The Director logs in. 2. Navigate to "Active Applications". 3. View the list of active patent applications. 4. The Director selects an application to view details.
Postconditions	The Director views detailed information of selected applications.
Alternate Flow	If there are no active applications, the system displays a message stating no applications are available.
Sub Flow	1. The Director filters applications based on topic or field. 2. Navigates between various applications using navigation options.
Global Alternate Flow	The Director cancels the view or exits to the main dashboard.

Use Case 12: Receive Confirmation from the Attorney about Patentability

Use Case ID	12
Use Case Name	Receive Confirmation from the Attorney about Patentability
Description	The PCC Admin receives a confirmation from the attorney about the patentability of the application and takes the necessary action.
Actor	PCC_Admin, Applicant
Precondition	The application has been forwarded to the attorney for review, and the attorney has confirmed patentability.

Main Flow	1. The PCC Admin receives the confirmation from the attorney (externally). 2. The PCC Admin updates the application status in the system. 3. The PCC Admin notifies the applicant about the confirmation. 4. The PCC Admin proceeds with the next steps in the patent filing process.
Postconditions	The patent application status is updated, and the applicant is informed.
Alternate Flow	Confirmation Not Received: • If the PCC Admin does not receive the confirmation from the attorney within the expected time: ◦ The PCC Admin sends a follow-up request to the attorney to inquire about the status of the review. ◦ If no response is received after the follow-up, the PCC Admin escalates the issue to the appropriate authority or selects an alternate attorney to review the patent application. Patentability Rejected: • If the attorney rejects the patentability of the application: ◦ The PCC Admin updates the application status to "Rejected" or "Needs Rework." ◦ The system sends a notification to the applicant informing them about the rejection. ◦ The applicant can then either submit corrections or withdraw the application based on the rejection feedback.
Sub Flow	Attorney Communication Log: • The PCC Admin logs all communication with the attorney regarding the patentability confirmation within the system. ◦ This includes emails, messages, or notes of phone calls to ensure there is a record of communication in case of future audits or reviews. Confirmation Acknowledgement: • After receiving the confirmation from the attorney, the PCC Admin acknowledges the attorney's input in the system by marking the review process as complete. ◦ The system generates a timestamp indicating when the confirmation was received.
Global Alternate Flow	The confirmation is not received, and the PCC Admin follows up with the attorney.

3.3 Business Rules:

3.3.1. Patent Application Submission:

- Only registered users (institute members) can submit a patent application.
- The applicant must provide all required information in the patent application form (e.g., invention title, description, inventor details).
- Incomplete applications must be flagged and sent back to the applicant for revisions before being forwarded for review.

3.3.2. Patent Application Review:

- The PCC Office must review all patent applications within a predefined time frame.
- If a patent application is deemed incomplete, the system will notify the applicant to resubmit within a given period (e.g. 10 days).

3.3.3. Approval Workflow:

- The PCC Office can only forward applications that are marked as complete.
- The Director must approve or reject applications within a predefined time frame received from the PCC Office.

3.3.4. Attorney Selection and Allocation:

- The system will suggest a list of attorneys based on their workload and area of expertise.
- The PCC Office will have final authority to allocate an attorney from the suggested list.
- The applicant must be notified immediately once an attorney is allocated.

3.3.5. Patentability Determination:

- The attorney must confirm the patentability (design or product patent) within a predefined time frame.
- The application can only be sent back to the Director for final approval after the attorney's confirmation.
- If the patent is not viable, the attorney provides a valid reason and notifies both the PCC Office and the applicant.

3.3.6. Final Approval and Contract Creation:

- The Director must provide final approval after receiving the attorney's confirmation.
- Once approved, the attorney must draft the patent contract within a predefined time frame (e.g., 7 business days).

- The system must notify the applicant upon the completion of the patent contract.

3.3.7. Data Privacy and Confidentiality:

- All patent applications and communications must be stored securely, with access restricted to authorized personnel (PCC Office, Director, Attorney).
- The system must ensure that patent details are confidential and are not accessible to unauthorized users.

3.3.8. Audit Trail and Logging:

- The system must maintain an audit trail of all actions taken during the patent application process, including submissions, reviews, approvals.
- All decisions (approvals, rejections, attorney confirmations) must be logged with timestamps.

3.3.9. Attorney Workload Distribution:

- The system must ensure an even distribution of cases among available attorneys based on their current workload to avoid overburdening any single attorney.

3.3.10. Legal Compliance:

- The system must ensure that all patent filings adhere to the country's patent laws and guidelines.
- All patent contracts must comply with legal standards and include necessary clauses for both design and product patents.

3.3.11. System Notifications:

- The system must automatically notify applicants, PCC Office, attorneys, and the Director at each key stage of the process (submission, approval, allocation, patentability confirmation, contract completion).

3.3.12. Resubmission Rules:

- If an application is rejected at any stage, the applicant can revise and resubmit within a specified window (e.g., 30 days).
- An application can only be resubmitted twice. After the third rejection, the application will be permanently archived.

3.4 Submission and Approval Workflow

3.4.1. Form Submission

Applicants can fill out an online Intellectual Property Filing Form, providing details such as administrative, technical, and commercialization aspects. The form also includes sections for inventor details, invention novelty, utility, ownership (funding and contribution breakdown), and supporting documents.

3.4.2. Review by PCC Office

After submission, the form is forwarded to the PCC Office for an initial review. The PCC can approve, reject, or request modifications through an automated notification system.

3.4.3. Forwarding to Director for Approval

Once approved by the PCC, the form is sent to the Director for approval. The Director's decision (approval or rejection) is relayed back to the system.

3.4.4. Attorney Allocation

After the Director's approval, the system suggests a list of attorneys based on their current workload. The PCC Office selects an attorney from this list, and the applicant is informed of the assigned attorney via automated notifications. The applicant is expected to directly communicate with the attorney for further processes such as a patentability check.

3.4.5. Attorney Confirmation

The attorney reviews the application and determines if the patent is valid and whether it falls under design or product patent categories. The attorney confirms this with the applicant and updates the PCC Office.

3.4.6. Final Director Approval

Once the attorney confirms the patentability, the PCC Office forwards the application back to the Director for final approval. The Director's approval is required to proceed with the patent contract.

3.4.7. Patent Contract Creation

Upon final approval by the Director, the assigned attorney drafts the patent contract, and the PCC Office oversees its completion.

3.4.8. Notifications and Status Tracking

Automatic notifications are triggered at key stages:

- When the application is submitted.
- When PCC approves or rejects the application.
- When the attorney is assigned and attorney-related communications.
- When the final patent is filed.

Applicants can log into the ERP portal to track the progress of their application throughout all stages.

4. External Interface Requirements

4.1 User Interfaces

Design user interfaces (e.g., using Figma) for each actor interacting with the system. Provide screenshots of different actors serving relevant functionality/use cases.

4.2 Software Interfaces

List external APIs or APIs from other Fusion modules that you need to use in implementing any of your use cases. For instance, module might be using FTS APIs to implement some workflow or Notification APIs to generate and receive notifications.

4.3 Communications Interfaces

Detail how different components communicate within the module (e.g., messaging services, data exchange formats.)

5. Nonfunctional Requirements

5.1 Performance

- The system should efficiently support up to 1000 simultaneous users without delays.
- Actions such as form submissions and status updates should be processed within 2 seconds to ensure a smooth user experience.

5.2 Security

- All sensitive information, including personal details and patent descriptions, must be securely encrypted both while stored (at rest) and while being transmitted (in transit).
- Role-based access control (RBAC) should ensure that only authorised users, such as PCC members and attorneys, can access specific sections of the system.

5.3 Usability

- The interface should be intuitive and user-friendly, so even users with little technical knowledge can easily submit patent applications.
- Each form field should have clear instructions or tooltips to guide users step-by-step during the application process.

5.4 Reliability

- The system should guarantee 99.9% uptime, ensuring that users can access the platform with minimal interruptions.
- Data backups must be conducted regularly to prevent data loss and ensure system reliability.

5.5 Scalability

- The system must be designed to easily accommodate more users and applications as demand increases.
- It should also support future integrations with external databases (e.g., patent databases) and include the ability to add new user roles as needed.

5.6 Compatibility

- The platform must function seamlessly across various web browsers (Chrome, Firefox, Safari) and devices (desktops, tablets, smartphones) to offer a consistent user experience.

6. Conclusion

The Patent Management Module for Fusion aims to automate and simplify the patent filing process for faculty and students. The system will ensure transparency, trackability, and timely notifications, making the entire process more efficient.